

Module 4
Chapter 2
Probability

$P(\text{Rain})$
 $Pr(\text{Rain})$

What is Probability?

Many events can't be predicted with total certainty. The best we can say is how *likely* they are to happen, using the idea of probability.

A probability reflects a degree of certainty about a statement between **0 (certain it's not true)** and **1 (certain it's true)**, inclusive. It often reflects the extent to which something is probable; the likelihood of something happening or being the case.

We can also use percentages to communicate probabilities, like 1% for a probability of 0.01, but in general the number between 0 and 1 behaves more nicely mathematically.

We can abbreviate a statement about a particular probability by saying something like $Pr(\text{Rain}) = 0.63$, where **Rain** indicates the statement "*it will rain today.*" We could also just abbreviate this as $P(\text{Rain})$, without the little r.

Since there are only **two possibilities**, it rains today or it doesn't, our belief **that it will not rain** must be $1 - 0.63 = 0.37$. This demonstrates a more general principle that the **probabilities of all possible disjoint (non-overlapping) outcomes must sum to 1**.

How do we get these probabilities?

To be certain of a probability, we need to employ **counting**. (Incidentally, that is true even for our weather example - the chance of rain in a forecast comes from counting how many simulations resulted in rain.)

$$P(X=5) = \frac{1}{6}$$

Counting to get Probabilities:

In general,

$$\text{Probability of an event happening} = \frac{\text{Number of ways it can happen}}{\text{Total number of outcomes}}$$

Probabilities are easiest to calculate when you can count the outcomes where an event is true and count the total possible number of outcomes. In addition, you need *some certainty* that the *outcomes are all equally likely*. Then the probability of your event is just:

$$\Pr(E) = \frac{|E|}{|S|}$$

Let's look at an example:

What are the chances of rolling a 5 with a die?

Well, there are 6 possible outcomes: 1, 2, 3, 4, 5, 6 and each outcome is equally likely. There is only 1 possible way of getting a 5 on a die roll: if we roll the die and it falls on the 5 face up!

$$\text{So, } \Pr(X=5) = \frac{1}{6}$$

We see the notation for the probability of getting a 5 and it has a new variable “X” added to it. This leads us to understanding *Random Variables*.

Random Variable

The random variable is associated with a *set of possible outcomes*, as well as a *distribution of probabilities* that gives the probability of each item in the set. For example, a Die could be a random variable with set of outcomes $\{1,2,3,4,5,6\}$ and a distribution that gives $1/6$ probability to all of the outcomes. Talking about random variables will later give us the chance to calculate the expectation of the random variable, or what its value is on average over time, as well as the variance, or how different our actual result tends to be from the expectation.

We will revisit random variables when we discuss about probability distributions!

$$P(H \text{ and } H \text{ and } H) = P(H) \times P(H) \times P(H) \\ = 1/2 \times 1/2 \times 1/2$$

Special words in Probability

- **Experiment:** A repeatable process with a set of possible outcomes. E.g. *“Throwing a Die”*
- **Outcome:** A possible result of an experiment. E.g. *“4 is an outcome of throwing a die”*
- **Trial:** A single performance of an experiment. E.g. *“I performed a coin toss experiment. After 4 trials I got 3 Heads and 1 Tail”*
- **Sample space:** All possible outcomes of an experiment. E.g. *Sample space for a single coin toss experiment: {H,T}*
- **Sample point:** One of the possible outcomes. E.g. *5 of clubs is a sample point for “choosing a card from a deck of cards” experiment.*
- **Event:** One or more of the outcomes of an experiment. An event can include just *one outcome* (e.g. Getting a head when tossing a coin) or can include *more than one outcome* (e.g. Rolling an even number: 2,4 or 6, when rolling a die).

Independent events

We are often interested in probabilities of two or more events happening at the same time. When *one event does not influence the occurrence of another event*, then they are called **independent events**. A good example will be coin tosses:

When we toss a coin multiple times, the previous outcome does not influence the outcome of the next coin toss. For two independent events A and B, the probability of both happening at the same time is given by:

$$\Pr(A \text{ and } B) = \Pr(A) \times \Pr(B)$$

Let's look at an example

What is the probability of getting three heads in a row in an experiment of coin tosses?

For each coin toss, the probability of getting a Head is 0.5. Let this be represented by $\Pr(H)$. We are interested in finding:

$$\Pr(H \text{ and } H \text{ and } H) = \Pr(H) \times \Pr(H) \times \Pr(H) = 0.5 \times 0.5 \times 0.5 = 0.125$$

Note the difference between the above question and this one: “We got 2 heads in a row. What is the probability of getting a Head in the third toss?”

Here we are just interested in the *third toss* and for every toss, $\Pr(H) = 0.5$

So, our answer will be 0.5!

Permutation/Combination for Probabilities

We have already established that probability requires counting: Counting the number of ways the event can happen and the total number of outcomes. We learnt in the previous chapter that counting can be done with or without order. Depending on the context, we use *Permutation/Combination to count the number of ways the event can happen as well as the total number of outcomes!*

Let’s look at an example

Card-drawing problems assume that all sets of cards you could draw have the same probability, but the ordering of the cards doesn’t matter. What’s the probability that if you draw 2 cards from a 52-card deck, they will both be aces (of which there are 4)?

**** A deck of cards contains:**

- 4 Aces
- 13 Hearts (Red)
- 13 Diamonds (Red)
- 13 Clubs (Black)
- 13 Spades (Black)
- 12 Face cards: 1 King, Queen and Jack for each shape **

We are interested in

$$P(X = 2 \text{ Aces}) = \frac{{}^4C_2}{{}^{52}C_2}$$

$$\frac{4}{52} \times \frac{3}{51}$$

of ways we get 2 Aces 4C_2

of ways of getting 2 cards ${}^{52}C_2$

$$\Pr(A = 2) = \frac{\text{Number of ways we can get two aces}}{\text{Number of ways we can pick any two cards}}$$

Number of ways we can get two aces:

$$C(4, 2) = \frac{4!}{(4 - 2)! 2!} = 6$$

Number of ways we can pick any two cards from a deck of 52 cards:

$$C(52, 2) = \frac{52!}{(52 - 2)! 2!} = 1326$$

$$\Pr(A = 2) = \frac{6}{1326} = 0.0045$$

Let's Practice

- What is the probability that a hand contains all 4 Aces, from a deck of 52 cards?
- In a random shuffling of n books, what is the chance that they end up in Alphabetical order? (no two books have the same name)
- A 4-digit pin is generated at random. What is the chance that it is between 1900 and 2021 (included)?
- What is the chance of drawing exactly one Ace in a 5-card hand?
- What is the chance of drawing no Aces in a 5-card hand?
- What is the probability of drawing at least one Ace in a 5-card hand?
- A game show has a spinner that can land in one of the 4 regions: blue, red, yellow or green. What is the probability of a contestant spinning three times and landing in the blue region each time?
- Jess has a sock drawer that contains 2 white socks, 2 brown socks and 4 black socks. What is the probability that she can randomly pick 2 socks and get a matching pair of black socks?

1. # of ways we can get 4 Aces: $4C_4$
~~#~~ # of ways we can get 1 remaining card: $48C_1$
~~#~~ # of ways we can pick a hand (5-cards) from
 52 cards: $52C_5$

$$\therefore P(\text{Hand of 4 Aces}) = \frac{4C_4 \times 48C_1}{52C_5} = 0.000018$$

$$\text{OR} \rightarrow \left(\frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \times \frac{1}{49} \times \frac{48}{48} \right) \times 5^{52C_5}$$

2. # ways they can be in alphabetical order: 1
~~#~~ # of ways we can arrange n books: $n!$

$$\therefore P(X = \text{alphabetical}) = \frac{1}{n!}$$

3. # of ways one can generate a 4-digit pin:
~~#~~ # of digits possible: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
 $= 10$
- $\rightarrow \overline{10} \times \overline{10} \times \overline{10} \times \overline{10} = 10000$

$$\begin{aligned} \# \text{ of ways it's a number between 1900 to 2021} \\ &= 2021 - 1900 + 1 \\ &= 121 + 1 = 122 \quad (\because \text{the years are inclusive}) \end{aligned}$$

$$\therefore P_{\text{H}}(\text{year b/w 1900 and 2021}) = \frac{122}{10000} = 0.0122$$

4. # of ways of getting exactly 1 Ace and 4 Non-Aces

$$4C_1 \times 48C_4$$

of ways of picking 5 cards from 52 cards:

$$52C_5$$

$$\therefore \text{Probability (1 Ace Hand)} = \frac{4C_1 \times 48C_4}{52C_5} = 0.299$$

5. # of ways of picking up No aces =

of ways of picking up 5 - Non aces:

$$48C_5$$

of ways of picking 5 cards from 52 cards:

$$52C_5$$

$$\therefore \text{Prob. (No Aces)} = \frac{48C_5}{52C_5} = 0.659$$

6. Prob of drawing at least 1 Ace:

$$1 - P(\text{No Aces}) = 1 - 0.659 = 0.341$$

OR

$$P(1 \text{ Ace}) + P(2 \text{ Aces}) + P(3 \text{ Aces}) + P(4 \text{ Aces})$$

7. $P(\text{Blue AND Blue AND Blue})$

$$= P(\text{Blue}) \times P(\text{Blue}) \times P(\text{Blue})$$

$$= \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{4^3} = \frac{1}{64}$$

8. No. of ways of picking 2 black socks:

$$4C_2$$

No. of ways of picking any 2 socks

$$8C_2$$

$$\therefore P(2 \text{ Black socks}) = \frac{4C_2}{8C_2} = \frac{3}{14}$$

Extra: What is the probability of picking a matching pair of socks if she picks up 2 socks

$$\rightarrow \frac{4C_2 + 2C_2 + 2C_2}{8C_2} = \frac{4C_2 + 2}{8C_2} = \frac{2}{7}$$

Binomial Distribution

Many a times we deal with experiments which have only *two possible outcomes*. E.g. when we toss a coin, it can be either head or tail, a person can either be employed or unemployed, an item can be defective or not defective etc. These outcomes are often labeled as “*success*” or “*failure*”. Remember that there is no connotation of goodness here!

Binomial Distribution is a discrete probability distribution which is used to represent such cases. Let's first talk about the notations we will be using to understand Binomial distribution:

- p : Probability of success
- q : Probability of failure; $q = 1 - p$
- **Bernoulli trial** : each repetition of an experiment involving only two outcomes
- n : Number of Bernoulli trials
- r : Number of successes

$$p + q = 1$$

We are often interested in the result of independent, repeated **Bernoulli trials**, i.e. the number of successes in repeated trials.

- **Independent** - the result of one trial does not affect the result of another trial.
- **Repeated** - conditions are the same for each trial, i.e. p and q remain constant across trials.

A binomial distribution gives us the probabilities associated with independent, repeated Bernoulli trials. In a binomial distribution the **probabilities of interest** are those of **receiving a certain number of successes**, r , in n independent trials each having only two possible outcomes and the same probability, p , of success. So, for

$$n = 10 \quad \phi = P(H) = 1/2$$

$$x = 4 \quad q = 1 - 1/2 = 1/2$$

example, using a binomial distribution, we can determine the probability of getting 4 heads in 10 coin tosses.

Let's take an example

$$\frac{HHHH TTTT}{P(H)^4 \times P(T)^6}$$

$$\begin{matrix} 4 & 6 \\ \phi & q \end{matrix}$$

Let's determine the probability of getting 4 heads in 10 coin tosses using Binomial distribution.

This is a 2-part process:

$$\begin{matrix} 10 - 4 \\ \phi & q \end{matrix}$$

- Determine the probability of one possible way the event of interest can occur
- Determine the number of ways the event of interest can occur

$$Pr(Event) = \text{Number of ways event can occur} \times Pr(\text{one Occurance})$$

Since we are interested in getting **4 Heads in 10 tosses**, getting a **Head** is our “**success**”. So, we have:

$$\begin{matrix} 10C_4 \\ nC_r \end{matrix}$$

$$n = 10$$

$$r = 4$$

$$p = 0.5$$

$$q = 1 - 0.5 = 0.5$$

Step 1:

Determine the probability of one possible way the event of interest can occur:

One way we can have 4 Heads in 10 tosses is, if our first four outcomes are Heads and the rest are Tails i.e. **SSSS FFFFFF**

The probability of this occurring:

$$Pr(S) \times Pr(S) \times Pr(s) \times Pr(S) \times Pr(F) \times Pr(F) \times Pr(F) \times Pr(F) \\ \times Pr(F) \times Pr(F)$$

The above is the probability of 4 i.e. **n successes** and 6 i.e. **(n - r) failures**. This can be written as:

$$\phi^4 q^6 \rightarrow \phi^r q^{(n-r)} \\ \rightarrow \left[\frac{n!}{r!(n-r)!} \right] p^r (1-p)^{(n-r)}$$

$$p^r q^{(n-r)} = p^r (1-p)^{(n-r)}$$

So, in this case the probability of getting 4 Heads in a row followed by 6 Tails will be: $.5^4 \times .5^6 = 0.0009765625$

$$n C_r$$

$$10 C_4$$

Step 2:

Determine the number of ways the event of interest can occur

Of course, this is just one of many ways that you can get 4 heads; further, because the repeated trials are all independent and identically distributed, each way of getting 4 heads is equally likely, e.g. the sequence S S S S F F F F F F is just as likely as the sequence S F S F F S F F S F. So, we also need to know ***how many different combinations produce 4 heads***, which we now know how to calculate (remember the combination formula?!)

So, putting both the steps together, we can now write the Binomial Distribution Formula:

In sampling from a stationary Bernoulli process, with the probability of success equal to p , the probability of observing exactly r successes in n independent trials is given by:

$$Pr(X = r; n, p) = \underbrace{C(n, r)}_{n C_r} p^r \underbrace{(1-p)^{(n-r)}}_q = \frac{n!}{r! (n-r)!} p^r (1-p)^{(n-r)}$$

Let's Practice

- A biology exam consists of 9 True/False questions. Meg forgot to study and guessed randomly on each question. What is the probability that she will answer exactly 4 questions correctly?
- A six-sided die is rolled 10 times. What is the probability that it will come up with an even number exactly 5 times?
- A manufacturer of Halogen bulbs knows that exactly 3% of the production of their 100W bulbs will be defective. What is the probability that exactly 5 bulbs in a carton of 144 bulbs will be defective?

- Your basketball team is playing a series of 5 games against your opponent. The winner is the team that wins more than 3 games out of 5. Let's assume that your team has 75% chances of winning a game. What is the probability that your team wins at least 3 games?

1. $n = 9$

$x = 4$

$p = 0.5$

$q = 0.5$

(since nothing is given, we assume $p = q$)

$$P(X=4; 9, 0.5) = {}^9C_4 (0.5)^4 (0.5)^{9-4}$$

2. $n = 10$

$x = 5$

$p = \text{prob. of even number} = \frac{3}{6} = \frac{1}{2}$

$q = \frac{1}{2}$

$$\therefore P(X=5; 10, \frac{1}{2}) = {}^{10}C_5 (0.5)^5 (0.5)^5$$

3. $n = 144$

$\mu = 5$

$p = 3\%$

$q = 97\%$

$$\therefore P(X=5; 144, 0.03) = {}^{144}C_5 (0.03)^5 (0.97)^{139}$$

$$= 0.169$$

4. $n = 5$

$p = 0.75$

$q = 0.25$

Case 1:

$\mu = 3$

Case 2:

$\mu = 4$

Case 3:

$\mu = 5$

$$P(X=3; 5, 0.75) + P(X=4; 5, 0.75) + P(X=5; 5, 0.75)$$

$$= {}^5C_3 (0.75)^3 (0.25)^2 + {}^5C_4 (0.75)^4 (0.25) + {}^5C_5 (0.75)^5$$

Extra Practice

1. A fair coin ($p=0.5$) is tossed until 3 Heads are obtained. What is the probability that exactly 3 Heads are obtained within the first 4 tosses?
2. For a process with $p=0.55$, what is the probability that you HAVE NOT achieved 6 successes after 10 trials?
3. Two cards are drawn at random. Find the probability that both are from the same suit.
4. If 3 people are chosen at random, find the probability that at least 2 share the same birthday. (Ignore leap years)

$$1. \quad 4C_3 (0.5)^3 (0.5)^1$$

$$2. \quad \text{We want } P(X \leq 5) \rightarrow$$

$$\begin{aligned} & {}^{10}C_0 p^0 (1-p)^{10} + {}^{10}C_1 p^1 (1-p)^9 + {}^{10}C_2 p^2 (1-p)^8 \\ & + {}^{10}C_3 p^3 (1-p)^7 + {}^{10}C_4 p^4 (1-p)^6 + {}^{10}C_5 p^5 (1-p)^5 \end{aligned}$$

$$\text{where } p = 0.55$$

$$\therefore P(X \leq 5) = \sum_{k=0}^5 {}^{10}C_k (0.55)^k (0.45)^{10-k}$$

3. Each suit has 13 cards. There are 4 suits

$$\therefore P(2 \text{ cards from same suit}) = \frac{4 \times {}^{13}C_2}{{}^{52}C_2} = \frac{4 \times 78}{1326}$$

$= 4/17$

4. # of ways in which none of the 3 share the same birthday

$$= 365 \times 364 \times 363$$

$$\# \text{ of possible birthdays} = 365 \times 364 \times 365$$

$$\therefore P(\text{at least 2 out of 3 have same birthday}) = 1 - \frac{365 \times 364 \times 363}{365 \times 365 \times 365} \approx 0.0082$$